0007. Reverse Integer

dium | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/rev

Problem

- Reverse the digits of a 32-bit signed integer. Return 0 if the result overflows. You may not use 64-bit integers.

Intuition

- Pop digits with % 10 and push them with res 10 + d.
- Before pushing, check whether res 10 + d would leave the int range.

Brute Force

- Use a long or a string reversal: forbidden by the 32-bit constraint.

Optimal Approach

- While x != 0: d = x % 10; x /= 10.
- If res > MAX/10, or res == MAX/10 and d > 7, return 0; mirror the check for MIN with -8.
- Res = res 10 + d.

Worked Example

```
x = -123.

d=-3 res=-3
d=-2 res=-32
d=-1 res=-321
```

Java Solution

```java
public class ReverseInteger {
    public int reverse(int x) {
        int res = 0;
        while (x != 0) {
            int d = x % 10;
            x /= 10;
            if (res > Integer.MAX_VALUE / 10 || (res == Integer.MAX_VALUE / 10 && d > 7)) return 0;
            if (res < Integer.MIN_VALUE / 10 || (res == Integer.MIN_VALUE / 10 && d < -8)) return 0;
            res = res * 10 + d;
        }
        return res;
    }
}
```

Complexity

- Time: O(log $|x|$). Space: O(1).

Pitfalls

- Java % keeps the sign of x, so negatives need no special handling.